

BI3052 Study Design Course Outline

Weeks (Uke) 35-49, høst (Fall) 2026:

Wednesday 10:15-14:00 room GL-VE VE22-SMASH and **Friday 12:15-16:00** in room GL-HB 316

Optional Course book: 'Experimental Design for the Life Sciences' Ruxton & Colegrave (3rd/4th edition, OUP, 2011/2016)

Other plus additional literature provided during the course in the Canvas BI3052 Learning Materials Folder

Schedule

Friday Aug 28th week 35

1. *Lecture: Introduction –Introduction to the scientific process - why experimental design is important; Introduction to Group Work (Cameron)*

Discussions – what are the important issues in your area?

Read Chap 1

Read: Schwartz "The Importance of stupidity in scientific research"

Wednesday Sep 2nd week 36

NO CLASS – SELF STUDY

Friday Sep 4th week 36

NO CLASS – SELF STUDY

Read Chap 2, 3, 4

Wednesday Sep 9th week 37

2. *Discussion of Importance of stupidity paper (Cameron)*

Group Presentations

Friday Sep 11th week 37

3. *Lecture: hypotheses, predictions, assumptions; generating alternative hypotheses and how to test them (Cameron) – Group Work*

Read: Wolff & Krebs: "Hypothesis testing and the scientific method revisited."

Wednesday Sep 16th week 38

4. *Group Presentations – generating and testing multiple alternative hypotheses and their predictions (Cameron)*

Friday Sep 18th week 38

5. *Lecture: Testing hypotheses using experimental vs non-experimental approaches, lab vs field studies (Cameron)*

Wednesday Sep 23rd week 39

6. Group Presentations – incorporating complexity into study design (Daniela).

Friday Sep 25th week 39

7. Confounding effects and controls in observations and experiments.

Carry-over effects, within- vs between-subjects comparisons... (Daniela)

Wednesday Sep 30th week 40

8. Group Presentations– recognizing sources of variation (Daniela)

Friday Oct 2nd week 40

Read Chap 5

9. Lecture: Replication, randomization, sampling, pseudoreplication, common environment & relatedness effects, and levels of analysis (Cameron)

Wednesday Oct 7th week 41

10. Group Presentation – Identifying the sampling unit and avoiding pseudoreplication (Daniela)

Friday Oct 9th week 41

11. Observer reliabilities, biases, and disturbance effects; repeatability, calibration, inaccuracy, and imprecision. Logistical trade-offs in data collection (Daniela)

Wednesday Oct 14th week 42

12. Group Presentations (Daniela)

Friday Oct 16th week 42

13. Introduction to R (Sigurd)

Descriptive and inferential statistics, parametric statistical models,

Read Chap 6-10

Wednesday Oct 21st week 43

NO CLASS SELF STUDY

Friday Oct 23rd week 43

14. basic principle of hypothesis testing, models with categorical predictor variables (Sigurd)

Wednesday Oct 28th week 44

NO CLASS SELF STUDY

Friday Oct 30th week 44

15. Models with continuous predictor variables, model assumptions, mixed models (Sigurd)

Wednesday Nov 4th week 45

16. Models with more than one categorical predictor variable, additive effects, interaction effects, models with both continuous and categorical predictor variables (Sigurd)

Friday Nov 6th week 45

17. Student presentations on study design, design options to address a simple hypothesis, linking statistical model to study design (Sigurd)

Wednesday Nov 11th week 46

18. More on linking statistical models to study design (Sigurd)

Friday Nov 13th week 46

19. Student presentations on study design, analysis of case study from A to Z (Sigurd)

Wednesday Nov 18th week 47

20. Question and answer session prior to Oral Exam

Friday Nov 20th week 48. Final Exam(6x30mins)

ORAL EXAM – Groups of 6 (randomly chosen) 30min pass/fail.